

Spectral Density Estimation for Stationary Processes: AR(p) Density Approximation and Nonparametric Smoothing

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ABSTRACT

Macroeconomic time series often exhibit macroeconomic cycles, seasonal patterns, and persistent long-run trends mixed with short-term fluctuations that are highly relevant for economic analysis and policy making. These components are often difficult to identify in the time domain, but the frequency domain provides a clearer representation by separating variance across different scales and highlighting persistent cycles, dominant or weak structural frequencies, and other systematic patterns that may not be visually apparent in the raw series. Consequently, accurate estimation of the spectral density for a stationary process is essential. In this paper, we examine two distinct methodologies for spectral estimation: a nonparametric approach based on windowed periodograms and a parametric approach grounded in ARMA model representations. Our simulation studies validate both approaches and demonstrate that parametric method excels under correct model specification while nonparametric method provides flexibility through data-driven estimation, making it particularly valuable when theoretical models are unavailable or uncertain.

1. Introduction

A substantial literature on spectral density estimation has been developed, spanning both nonparametric and parametric approaches. The nonparametric approach to estimating spectral density offers several advantages, one of which is the flexibility since it does not require a specific model. Therefore, it is well suited for uncovering the underlying dependence pattern, which allows us to study the latent cycles, seasonal components, and other structural frequencies that may not be captured by low-order parametric models. However, these benefits come with important limitations. The periodogram is not a consistent estimator, its variance does not vanish asymptotically, which motivates the use of smoothing, tapering, or averaging procedures such as Welch's method (Welch (1967)) to obtain a stable estimate. Another approach in the nonparametric realm is smoothing of the log periodogram, as demonstrated in Wahba (1980). The log spectral density is obtained by applying a smoothing spline on the log of the periodogram, with the smoothing parameter λ as well as the degree of the smoothing spline chosen to minimize the estimate of the integrated

mean squared error. Recent advances in smoothing periodogram include Bayesian nonparametric approaches, such as in Choudhuri et al (2004). This paper explores constructing the prior using Bernstein polynomial, and updating the pseudoposterior distribution using Whittle likelihood, which results in consistent pseudoposterior distribution.

The parametric approach, typically based on ARMA models, provides a convenient and analytically tractable framework for spectral estimation. Because the spectral density of an ARMA process admits a closed-form expression, computation is straightforward and statistically efficient when the model appropriately reflects the underlying dynamics. Parametric estimators generally exhibit lower variance than their nonparametric counterparts and can perform well even with relatively short samples. Beyond ARMA models, several other parametric frameworks have been developed for spectral analysis. Granger and Joyeux (1980) introduced the ARFIMA model by generalizing the concept of differencing to fractional orders, showing that allowing the differencing parameter d to take non-integer values produces a stationary process with a parametric spectral density that decays as a power law at low frequencies, thereby capturing long-memory behavior that finite-order ARMA models cannot represent. Harvey and Trimbur (2003) extend the structural time-series framework by developing a state-space model in which the implied spectral density corresponds to a Butterworth-type bandpass filter. By estimating the model parameters via maximum likelihood in state-space form, the resulting Kalman smoother extracts business-cycle components through a model-implied parametric filter.

This report is organized as follows. Section 1 explores the parametric approach to spectral density estimation. We begin by establishing a theoretical result that demonstrates that the spectral density of any stationary process can be arbitrarily closely approximated by an $\text{AR}(p)$ spectral density under appropriate regularity conditions. This approximation approach is then assessed through simulation studies that examine both the goodness-of-fit of correctly specified models and the consequences of model misspecification. Section 2 addresses nonparametric spectral density estimation techniques, discussing computational methods leveraging circulant matrix structures and Welch’s method for smoothing estimator, with the latter evaluated through simulation studies. Section 3 compiles our findings from both parametric and nonparametric frameworks, highlights the fundamental trade-offs between efficiency and robustness in spectral density estimation, and outlines promising directions for future research.

2. Spectral Density Estimation: Parametric Approach

2.1. Spectral Density Approximation Using $\text{AR}(p)$ Processes

A central idea in the parametric approach to spectral density estimation is to approximate the underlying stationary process by a member of a parametric family whose spectral density admits a closed-form expression. Among the most widely used families are autoregressive (AR) models, which allows for transparent analysis and straightforward estimation.

Definition 2.1. A stochastic process $\{X_t\}$ is said to be **weakly stationary or wide sense stationary (wss)** if:

- (a) $\mathbb{E}[X_t] = \mu$,
- (b) $\text{Var}(X_t) = \sigma^2$,
- (c) $\gamma_X(t, s) = \gamma_X(t - s)$.

Weak stationarity ensures that the second-order structure is time-invariant and that a spectral density exists.

Definition 2.2. An **autoregressive process of order p** , denoted $AR(p)$, is a stochastic process $\{Y_t\}$ satisfying

$$Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + Z_t$$

where $Z_t \sim WN(0, \sigma^2)$ for all $t, s \in T$

Let $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$ denote the characteristic AR polynomial. If all roots of $\phi(z)$ lie outside the unit circle, then the $AR(p)$ process is causal, weakly stationary, and possesses a spectral density that can be derived as a special case of the $ARMA(p, q)$ result below.

Theorem 2.1: Spectral Density of the $ARMA(p, q)$ Process. Let $\{Y_t\}$ be an $ARMA(p, q)$ process, not necessarily causal or invertible, satisfying

$$\phi(B)Y_t = \theta(B)\varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2) \quad (1)$$

where the characteristic AR and MA polynomials $\phi(\zeta)$ and $\theta(\zeta)$ have no common zeros, and $\phi(\zeta)$ has no zeros on the unit circle. Then $\{Y_t\}$ has spectral density

$$f_Y(\lambda) = \frac{\sigma^2}{2\pi} \frac{|\theta(e^{-i\lambda})|^2}{|\phi(e^{-i\lambda})|^2}, \quad \lambda \in [-\pi, \pi]. \quad (2)$$

Applying Theorem 2.1 to the $AR(p)$ case (i.e., setting $\theta(e^{-i\lambda}) \equiv 1$), the spectral density of an $AR(p)$ process is

$$\begin{aligned} f_p(\lambda) &= \frac{\sigma^2}{2\pi} |\phi(e^{-i\lambda})|^{-2} \\ &= \sigma^2 (2\pi)^{-1} |\phi(e^{-i\lambda})|^{-2} \\ &= A^{-1} |\phi(e^{-i\lambda})|^{-2} \end{aligned} \quad (3)$$

where $\phi(e^{-i\lambda}) = 1 - \phi_1 e^{-i\lambda} - \dots - \phi_p e^{-ip\lambda}$, $A \triangleq 2\pi\sigma^{-2}$.

Theorem 2.2: Trigonometric Weierstrass Approximation. Suppose f is a continuous real-valued function defined on the real interval $[a, b]$. For every $\epsilon > 0$, there exists a polynomial p such that for all x in $[a, b]$, we have the supremum norm $\|f(x) - p(x)\| < \epsilon$

Theorem 2.3: Fejér–Riesz Factorization. *If $T_N(\lambda) = \sum_{k=-N}^N c_k e^{ik\lambda}$ is a nonnegative trigonometric polynomial, then there exist $C > 0$ and a polynomial $Q(z) = \sum_{j=0}^M q_j z^j$ such that*

$$T_N(\lambda) = C |Q(e^{-i\lambda})|^2, \quad \lambda \in [-\pi, \pi].$$

Theorem 2.4: Spectral Approximation. *For any stationary process with continuous spectral density f , the spectral density can be approximated arbitrarily well by that of a causal AR(p) process. That is, for any $\varepsilon > 0$ there exists p such that:*

$$\sup_{\lambda} |f(\lambda) - f_p(\lambda)| < \varepsilon.$$

Proof. In Eqn. 3, the AR(p) spectral density is $A^{-1}|\phi(e^{-i\lambda})|^{-2}$, and in order to approximate a stationary process $\{X_t\}$, we will show that there exists a small $\epsilon > 0$ such that

$$|[A|\phi(e^{-i\lambda})|^2]^{-1} - f(\lambda)| < \epsilon \quad \forall \lambda \in [-\pi, \pi] \quad (4)$$

$$\begin{aligned} |[A|\phi(e^{-i\lambda})|^2]^{-1} - f(\lambda)| &= |[A|\phi(e^{-i\lambda})|^2]^{-1} - f_p(\lambda) + f_p(\lambda) - f(\lambda)| \\ &\leq |[A|\phi(e^{-i\lambda})|^2]^{-1} - f_p(\lambda)| + |f_p(\lambda) - f(\lambda)|. \end{aligned} \quad (5)$$

where $f_p(\lambda) = \max(\frac{\epsilon}{2}, f(\lambda))$, since we will be approximating the reciprocal of $f_p(\lambda)$, this choice of $f_p(\lambda)$ ensures that its reciprocal will not blow up when it gets too small. We then have:

$$0 \leq |f_p(\lambda) - f(\lambda)| \leq \frac{\epsilon}{2}. \quad (6)$$

Let $M \triangleq \sup_{\lambda} f_p(\lambda)$, then we have the inequality:

$$\frac{\epsilon}{2} < f_p(\lambda) < M \iff \frac{1}{M} < \frac{1}{f_p(\lambda)} < \frac{2}{\epsilon}. \quad (7)$$

$f(\lambda)$ is positive and symmetric $f(\lambda) = f(-\lambda)$ on the interval $[-\pi, \pi]$, hence $f_p(\lambda)$ is also positive and symmetric on $[-\pi, \pi]$, $\frac{1}{f_p(\lambda)}$ is also positive and symmetric on a compact interval, therefore we can apply Theorem 2.2, so there exists a polynomial $P_m(\cos(\lambda)) = \sum_{k=0}^N c_k \cos(k\lambda)$ such that:

$$\left| P_m(\cos(\lambda)) - \frac{1}{f_p(\lambda)} \right| < \delta, \quad \text{where } \delta = \min\left(\frac{1}{2M}, \frac{\epsilon}{4M^2}\right). \quad (8)$$

In the Appendix, we derive a suitable δ . Next, using Euler's formula, we can rewrite our polynomial

$P_m(\cos(\lambda))$:

$$\begin{aligned}
 P_m(\cos(\lambda)) &= \sum_{k=0}^N c_k \cos(k\lambda) \\
 &= \sum_{k=0}^N c_k \frac{e^{ik\lambda} + e^{-ik\lambda}}{2} \\
 &= \sum_{k=0}^N c_k \frac{e^{ik\lambda}}{2} + \sum_{k=0}^N c_k \frac{e^{-ik\lambda}}{2} \\
 &= \sum_{k=0}^N c_k \frac{e^{ik\lambda}}{2} + \sum_{k=-N}^0 c_{-k} \frac{e^{ik\lambda}}{2} \\
 &= \sum_{k=1}^N c_k \frac{e^{ik\lambda}}{2} + \sum_{k=-N}^{-1} c_k \frac{e^{ik\lambda}}{2} + c_0
 \end{aligned} \tag{9}$$

Let $d_0 = c_0, d_k = \frac{c_k}{2}$

$$= \sum_{k=-N}^N d_k e^{ik\lambda} \triangleq g(\lambda);$$

Substitute the result in Eqn. 9 into Eqn. 8, we have:

$$\left| g(\lambda) - \frac{1}{f_p(\lambda)} \right| < \delta \tag{10}$$

$$\begin{aligned}
 g(\lambda) &\geq \frac{1}{f_p(\lambda)} - \delta \\
 \delta &\leq \frac{1}{2M} \leq \frac{1}{2f_p(\lambda)} \text{ (follows from equation (7))} \\
 \implies g(\lambda) &\geq \frac{1}{f_p(\lambda)} - \frac{1}{2f_p(\lambda)} \geq 0.
 \end{aligned} \tag{11}$$

Therefore, we can apply Theorem 2.3 to the non-negative and symmetric trigonometric polynomial $g(\lambda)$, and we can find $\phi(e^{-i\lambda}) = 1 + \phi_1 e^{-i\lambda} + \dots + \phi_p e^{-i\lambda p}$ such that $g(\lambda) = A|\phi(e^{-i\lambda})|^2$. Substituting this result into Eqn. 10, we have:

$$\left| [A|\phi(e^{-i\lambda})|^2] - \frac{1}{f_p(\lambda)} \right| \leq \delta, \tag{12}$$

Let $k(\lambda) \triangleq \frac{1}{f_p(\lambda)}$ and we have $g(\lambda) = [A|\phi(e^{-i\lambda})|^2]$, then Eqn. 12 becomes

$$|g(\lambda) - k(\lambda)| \leq \delta \tag{13}$$

We then have:

$$\begin{aligned}
 |[A|\phi(e^{-i\lambda})|^2]^{-1} - f_p(\lambda)| &= \left| \frac{1}{g(\lambda)} - \frac{1}{k(\lambda)} \right| \\
 &= \left| \frac{k(\lambda) - g(\lambda)}{g(\lambda)k(\lambda)} \right| \\
 &\leq \frac{\delta}{|g(\lambda)k(\lambda)|}
 \end{aligned} \tag{14}$$

From Eqn. 13, we have

$$\begin{aligned}
 |g(\lambda) - k(\lambda)| &\leq \delta \\
 \implies g(\lambda) &\geq k(\lambda) - \delta \\
 &= \frac{1}{f_p(\lambda)} - \delta \\
 &\geq \frac{1}{M} - \delta \\
 &\geq \frac{1}{2M},
 \end{aligned} \tag{15}$$

since

$$\delta = \min\left(\frac{1}{2M}, \frac{\epsilon}{4M^2}\right).$$

Since $g(\lambda) \geq 0$ and $k(\lambda) = \frac{1}{f_p(\lambda)} > 0$, we obtain

$$|g(\lambda)k(\lambda)| = g(\lambda)k(\lambda) > \frac{1}{2M} \times \frac{1}{M} = \frac{1}{2M^2}. \tag{16}$$

Substituting this result into Eqn. 14 gives

$$\begin{aligned}
 |[A|\phi(e^{-i\lambda})|^2]^{-1} - f_p(\lambda)| &\leq 2M^2\delta \\
 &\leq \frac{\epsilon}{2}.
 \end{aligned} \tag{17}$$

Combining Eqn. 6 and Eqn. 17, we obtain

$$|[A|\phi(e^{-i\lambda})|^2]^{-1} - f(\lambda)| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \tag{18}$$

This completes our proof. $\square$

2.2. Simulation Study: ARMA-Based Spectral Approximation of a Stationary Process

To illustrate the connection between the spectral approximation theorem and practical model fitting, we simulate data from a known AR(2) process and then apply a standard Box–Jenkins method to determine the model applicable to this data.

The sample ACF and PACF in Figure 1 indicates clear evidence of an autoregressive structure: the PACF exhibits two statistically significant spikes before cutting off, while the ACF displays a gradually decaying pattern. This behavior is consistent with an AR(2) model, therefore we move forward with using this model to approximate the spectral density.

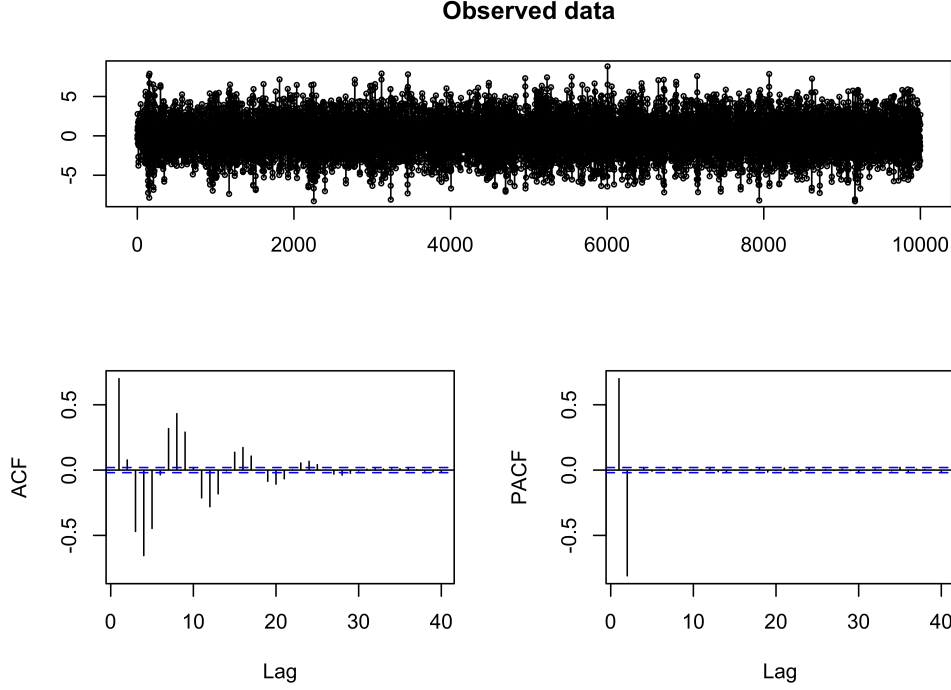


Fig. 1.— **ACF and PACF of the simulated series.** The simulated series are stationary, with exponential decay in ACF and two statistically significant spikes in PACF plot, which allows us to fit an AR(2) model to this series.

Using the Yule-Walker equations, we obtain the estimated coefficients $\hat{\phi}_1 = 1.266028$ and $\hat{\phi}_2 = -0.8091923$. Using the formulation in Eqn. 3, we obtain the estimated spectral density $\hat{F}_p(\lambda)$ for the AR(2) model, which can be observed from Figure 2.

To assess the goodness-of-fit between the true spectral distribution and the one implied by the fitted AR(2) model, we implement the CvM test. This test quantifies the discrepancy between two distribution functions by measuring the average squared distance between them. In our setting, we compare the empirical cumulative spectral distribution of the AR(2) estimator, $\hat{F}_{AR}(\omega)$, with the true cumulative spectral distribution $F(\omega)$. The hypotheses are

$$\begin{aligned} H_0 : \hat{F}_{AR}(\omega) &= F(\omega) \\ H_A : \hat{F}_{AR}(\omega) &\neq F(\omega) \end{aligned} \tag{19}$$

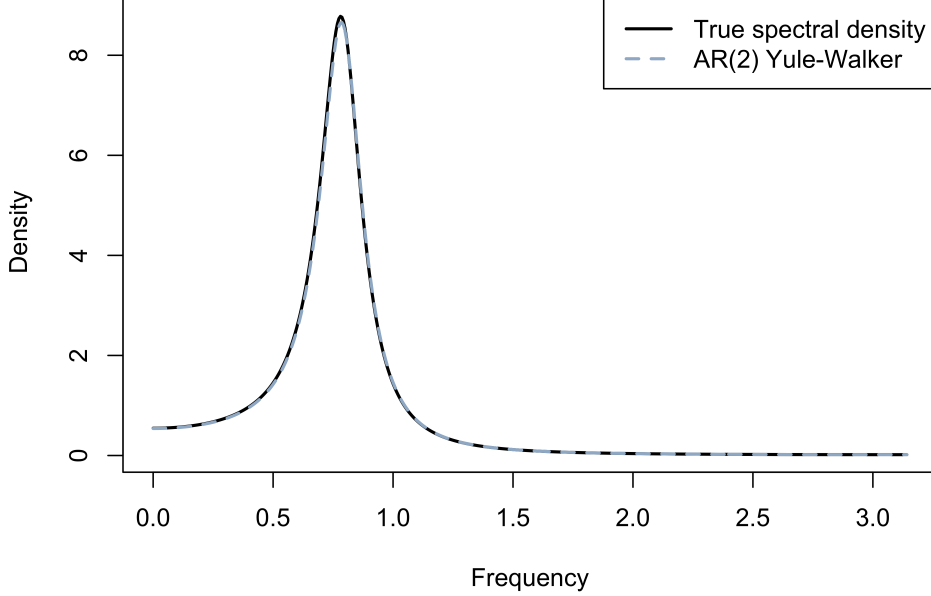


Fig. 2.— **True Spectral Density and $AR(2)$ Spectral Density.** The estimated spectral density obtained from fitting $AR(2)$ model on the simulated process is visually close to the true spectral density.

The CvM test statistic is then defined as

$$T_{CvM} = \int_{\omega_{min}}^{\omega_{max}} (\hat{F}_{AR}(\omega) - F(\omega))^2 d\omega. \quad (20)$$

For spectral density testing, we cannot use the standard CvM critical values since it requires i.i.d observations, but our time series has temporal dependency. Therefore, we are going to generate the null distribution via bootstrap, which gives the distribution of the test statistics under the null hypothesis. For each bootstrap sample, we sample an $AR(2)$ from the residuals of the fitted $AR(2)$ model, which resembles white noise as seen in Figure 3, and calculate the test statistics for each sample. The p-value used is the proportion of observing the bootstrapped test statistics that is greater than the observed test statistics. Our study suggests a p-value 0.706, which means that the estimated spectral density is consistent with the true spectral density at the significance level $\alpha = 0.05$.

To assess the extent of model misspecification, we compute the KL divergence between each candidate model’s spectral density and the true spectral density of the data-generating process. Specifically, we evaluate $MA(q)$, $AR(p)$ and $ARMA(p, q)$ models over a range of orders (p, q) as potential approximating models for the underlying stationary process. Because the data-generating

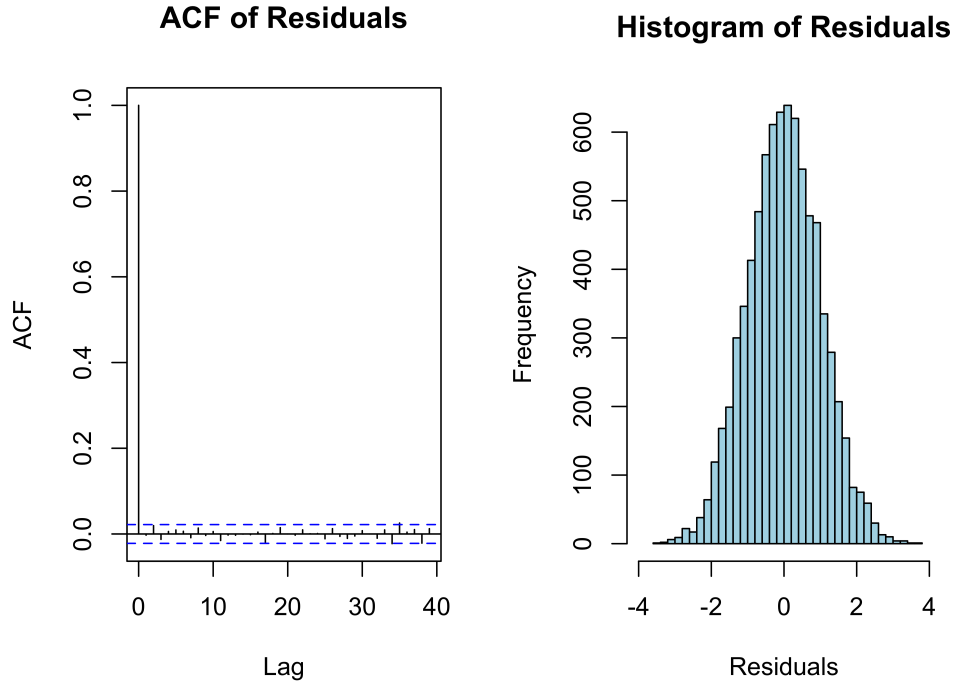


Fig. 3.— **Residuals of the $AR(2)$ model.** As observed from the ACF plot, the residuals seem to resemble white noise, and from the histogram, the residuals look normally distributed as well.

mechanism is a causal $AR(2)$, process—i.e., one whose autoregressive polynomial has all roots outside the unit circle—its spectral density admits an infinite-order moving average representation. Consequently, any finite-order $MA(q)$ with $q \leq 5$ cannot reproduce the true spectrum, and therefore exhibits a substantially larger KL divergence relative to the AR and ARMA models. Additionally, a lower order than the true order of the AR model also performs poorly because it cannot reproduce the correct pattern of the spectral density due to having too few roots for the polynomial. This pattern is clearly reflected in Figure 4, where all finite-order MA models display markedly higher divergence values. In contrast, any $ARMA(2, q)$ model performs comparably to the correctly specified $AR(2)$ model in estimating the spectral density of the underlying process, as the inclusion of additional moving-average terms does not alter the fundamental autoregressive structure that governs the spectrum. This reinforces the necessity of careful model identification (e.g., via a Box–Jenkins-type procedure) prior to estimating spectral quantities, as underspecified models can lead to significant distortions in the estimated spectrum.

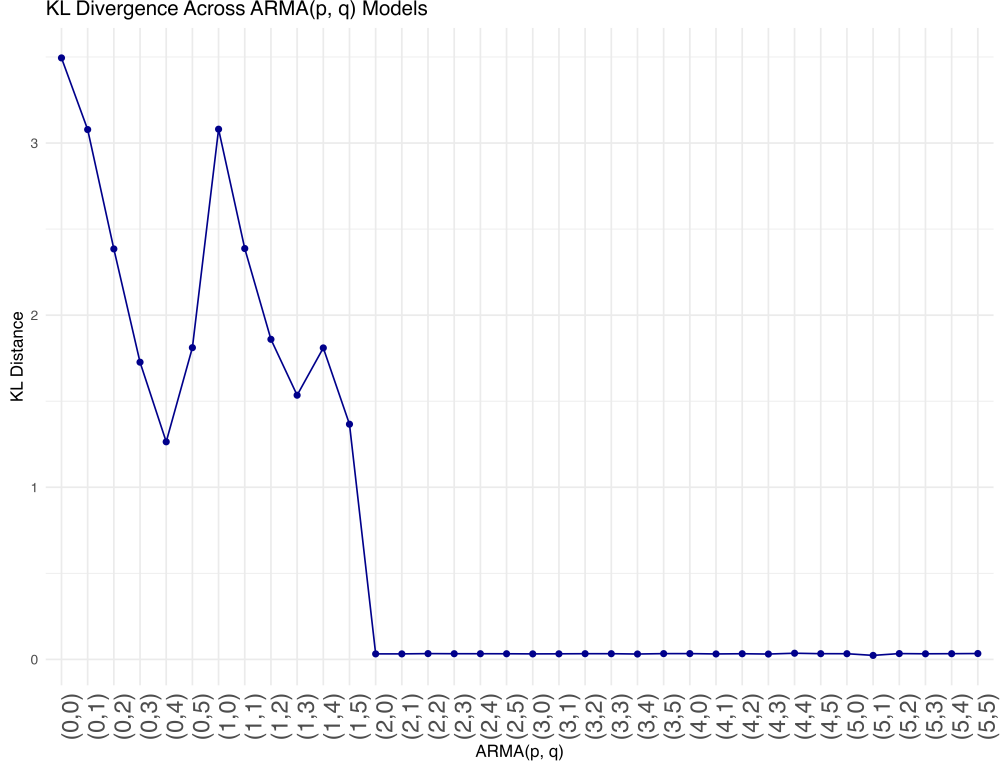


Fig. 4.— **KL divergence across different models.** KL divergence between the true $AR(2)$ spectral density and various AR, MA, and ARMA models. Finite-order MA models and low order $AR(p)$ and $ARMA(p, 1)$ ($p = 1$) models exhibit large divergence due to misspecification, whereas $AR(p)$ and $ARMA(p, q)$ with $p \geq 2$ models achieve near-zero divergence and successfully recover the underlying spectral structure.

2.3. Estimation of the Autocovariance Function (ACVF)

In this section, we will explore techniques to estimate the stationary process X_t without making a model assumption. Instead, we will estimate its ACVF $\hat{\Gamma}(k)$ and its spectral density $\hat{F}(\lambda)$. As suggested in Brockwell and Davis (2016), we use cross moment sampling to estimate the ACVF as follows:

$$\hat{\Gamma}_X(h) = \frac{1}{n} \sum_{t=1}^{n-|h|} (X_{t+|h|} - \bar{X}_n)(X_t - \bar{X}_n) \quad (21)$$

where the moment estimator of the mean μ of a stationary process is the sample mean: $\bar{X} = \frac{X_1 + \dots + X_n}{n}$.

Note that the estimator is biased, even if we scale it by $(n - h)^{-1}$ instead of n^{-1} .

Theorem 2.5: Central Limit Theorem for a Stationary Process (Thm 7.1.2 in Davis & Brockwell Time Series Theory and Methods). *Let $\{X_t\}_{t=1}^T$ be a strictly stationary, zero-*

mean, linear process:

$$X_t = \mu + \sum_{i=0}^{\infty} \psi_i Z_{t-i}$$

where $\sum_{i=0}^{\infty} |\psi_i| < \infty$, $Z_t \sim IID(0, \sigma^2)$.

Then:

$$\bar{X}_n \sim AN(\mu, n^{-1}\nu)$$

where $\nu = \sum_{h=-\infty}^{\infty} \gamma(h) = \sigma^2(\sum_{i=0}^{\infty} \psi_i)^2$, and $\gamma(\cdot)$ is the autocovariance function of $\{X_t\}$.

WLOG, we assume that $\mathbb{E}[X_t] = 0$ then we can simplify $\gamma(h)$ using Theorem 2.5

$$\begin{aligned} \mathbb{E}[\hat{\Gamma}_X(h)] &= \frac{1}{n} \sum_{t=1}^{n-|h|} \mathbb{E}[X_{t+|h|} X_t + \bar{X}_n^2] \\ &= \frac{1}{n} \sum_{t=1}^{n-|h|} \gamma(h) + \mathbb{E}[\bar{X}_n^2] \end{aligned}$$

Using Theorem 2.5 : $\sqrt{n} \bar{X}_n \xrightarrow{d} N(0, \nu)$, so that $\bar{X}_n = \mathcal{O}_p(n^{-1/2})$ (22)

$$\mathbb{E}[\bar{X}_n^2] = \text{Var}(\bar{X}_n) = \frac{\nu}{n} + \mathcal{O}\left(\frac{1}{n}\right).$$

Therefore:

$$\mathbb{E}[\hat{\Gamma}_X(h)] = \left(1 - \frac{|h|}{n}\right) \gamma(h) + \frac{\nu}{n} + \mathcal{O}\left(\frac{1}{n}\right)$$

This estimator is obviously biased, however, it becomes asymptotically unbiased as $n \rightarrow \infty$. We consider scaling the sampled cross moments with n^{-1} since it guaranties that the k -dimensional sample covariance matrix ($\hat{\Gamma}_k$) is positive definite, which is the property we need to ensure that the estimated spectral density $\hat{F}$ is a valid density. The k -dimensional sample covariance matrix is:

$$\hat{\Gamma}_k = \begin{bmatrix} \hat{\gamma}(0) & \hat{\gamma}(1) & \cdots & \hat{\gamma}(k-1) \\ \hat{\gamma}(1) & \hat{\gamma}(0) & \cdots & \hat{\gamma}(k-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\gamma}(k-1) & \hat{\gamma}(k-2) & \cdots & \hat{\gamma}(0) \end{bmatrix} \quad (23)$$

We can rewrite $\hat{\Gamma}_k$ as:

$$\hat{\Gamma}_k = \frac{1}{n} C^T C \quad (24)$$

Where C is an $n \times k$ matrix, in which each column represents the centered time series at different

lags:

$$C = \begin{bmatrix} X_1 - \bar{X} & X_2 - \bar{X} & X_3 - \bar{X} & \cdots & X_k - \bar{X} \\ X_2 - \bar{X} & X_3 - \bar{X} & X_4 - \bar{X} & \cdots & X_{k+1} - \bar{X} \\ X_3 - \bar{X} & X_4 - \bar{X} & X_5 - \bar{X} & \cdots & X_{k+2} - \bar{X} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_{n-k+1} - \bar{X} & X_{n-k+2} - \bar{X} & X_{n-k+3} - \bar{X} & \cdots & X_n - \bar{X} \end{bmatrix}$$

In which we can see that

$$\begin{aligned} a^T \hat{\Gamma}_k a &= a^T \frac{1}{n} C^T C a \\ &= \frac{1}{n} (a^T C^T) (C a) \\ &= \|C a\|^2 \geq 0 \end{aligned} \tag{25}$$

Therefore, the estimate $\hat{\Gamma}(k)$ from the Eqn. 21 is positive definite, hence by Theorem ACVF it is an autocovariance function, and the spectral density is the spectral distribution of both $\gamma(k)$ and the stationary process $\{X_t\}$.

When we scale by $(n - |k|)^{-1}$, not all lags are given equal weight, and the larger lags are given more weight, that is, $\tilde{\gamma}(k-1) = \frac{1}{n-k+1} \sum_{t=1}^{n-k+1} (X_t - \bar{X})(X_{t+k-1} - \bar{X})$. Hence, each element has a different denominator, so we cannot rewrite the matrix $\hat{\Gamma}_k$ as $D^T D$ for some matrix D . Therefore, $\hat{\Gamma}_k$ might not be positive definite.

2.4. Estimation of the Spectral Density

Given the estimated ACVF $\{\hat{\Gamma}(h)\}_{h=-(n-1)}^{n-1}$, there are several ways to construct an estimator $\hat{F}(\omega)$ of the spectral density. In this paper we focus on two broad classes: (i) estimators obtained by (finite) Fourier transforms of the estimated ACVF, and (ii) smoothed or lag-window type estimators. For each method, we will study its consistency.

2.4.1. Estimation of Spectral Density using Discrete Fourier Transform of Circulant Matrices

A natural ACVF-based estimator of the spectral density is the finite Fourier transform

$$\hat{F}(\omega) = \frac{1}{2\pi} \sum_{h=-(n-1)}^{n-1} \hat{\Gamma}(h) e^{-ih\omega} \tag{26}$$

The empirical covariance matrix $\hat{\Gamma}_n = \{\hat{\Gamma}(j-k)\}_{j,k=1}^n$ has a Toeplitz structure. For large n it is common to approximate this Toeplitz matrix by a circulant matrix whose first row is constructed from the ACVF. Circulant matrices are diagonalized by the discrete Fourier transform (DFT), so their eigenvalues are obtained by applying the DFT to the first row. This provides an efficient

$O(n \log n)$ implementation, via the fast Fourier transform (FFT), both for evaluating spectral estimates on a frequency grid and for performing large-scale covariance computations based on $\hat{\Gamma}_n$.

We start by constructing a circulant matrix C_n from the Toeplitz matrix in Eqn. 23, where $\sigma_0 = \hat{\Gamma}(0), \sigma_h = \hat{\Gamma}(h), 1 \leq h \leq n-1$:

$$[C_n]_{ij} = \sigma_{(i-j) \bmod n} \quad (27)$$

Because the modular index satisfies

$$(i-j) \bmod n = \begin{cases} h & \text{equivalent to lag } +h \\ n-h & \text{equivalent to lag } -h \end{cases} \quad (28)$$

Since the process is stationary $\gamma(h) = \gamma(-h)$, we can still use $\sigma_h = \hat{\Gamma}(h)$.

Therefore, we can safely define that the generator σ as follows:

$$\begin{aligned} \sigma_0 &= \hat{\Gamma}(0) \\ \sigma_h &= \gamma(\hat{h}), \quad h = 1, \dots, n-1 \\ \sigma_{n-h} &= \gamma(\hat{-h}), \quad h = 1, \dots, n-1 \end{aligned} \quad (29)$$

We can show that the Frobenius norm of the difference between the circulant matrix C_n and the ACVF matrix $\hat{\Gamma}_n$ converges to 0 as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \|\hat{\Gamma}_n - C_n\|_F^2 = 0 \quad (30)$$

The proof is covered in Prof. Peters Lecture Notes.

The circulant matrix $C_n \in \mathbb{C}^{n \times n}$ admits the spectral decomposition:

$$C_n = F^{-1} \Lambda F, \quad (31)$$

where F denotes the $n \times n$ discrete Fourier transform (DFT) matrix with entries

$$F_{jk} = \exp\left(-\frac{2\pi i}{n} jk\right), \quad j, k = 0, \dots, n-1. \quad (32)$$

The matrix $\Lambda = \text{diag}(\lambda_0, \dots, \lambda_{n-1})$ is diagonal and contains the eigenvalues of C , given explicitly by the DFT of its first column:

$$\lambda_k = \sum_{h=0}^{n-1} \sigma_h \exp\left(-\frac{2\pi i}{n} hk\right), \quad k = 0, \dots, n-1, \quad (33)$$

where $(\sigma_0, \dots, \sigma_{n-1})$ is the generating vector of the circulant matrix. Thus, the factorization $C = F^{-1} \Lambda F$ expresses the fact that every circulant matrix is unitarily diagonalized by the Fourier basis. Therefore, the spectral density embedded at the Fourier frequencies is:

$$\begin{aligned}
\hat{F}_{\text{circ}}(\omega_k) &= \frac{1}{2\pi} \lambda_k \\
&= \frac{1}{2\pi} \left(\hat{\Gamma}(0) + \sum_{h=1}^{n-1} \hat{\gamma}(h) e^{-ih\omega_k} + \sum_{h=1}^{n-1} \hat{\gamma}(-h) e^{-i(-h)\omega_k} \right) \\
&= \frac{1}{2\pi} \left(\hat{\Gamma}(0) + \sum_{h=1}^{n-1} \hat{\gamma}(h) e^{-ih\omega_k} + \sum_{h=-(n-1)}^{-1} \hat{\gamma}(h) e^{-i(h)\omega_k} \right) \\
&= \frac{1}{2\pi} \sum_{h=-(n-1)}^{n-1} \hat{\gamma}(h) e^{-ih\omega_k}, \quad \omega_k = \frac{2\pi k}{n}
\end{aligned} \tag{34}$$

To check for biasness, we compute the first moment of the estimator $\hat{F}_{\text{circ}}(\omega_k)$.

$$\mathbb{E}[\hat{F}_{\text{circ}}(\omega_k)] = \frac{1}{2\pi} \sum_{h=-(n-1)}^{n-1} \mathbb{E}[\hat{\Gamma}(h)] e^{-ih\omega_k} \tag{35}$$

This estimator is asymptotically unbiased because from Eqn. 22, we can see that $\hat{\Gamma}(h)$ is asymptotically unbiased.

However, this estimator is inconsistent. Therefore, we will show another way to estimate the spectral density using Welch's method, which is an average of successive blocks of periodogram over time. This method will reduce the variance of the raw periodogram above.

2.4.2. Welch's Method for Spectral Density Estimation

Estimation Procedure

We begin with a stationary time series data $\{X_t\}_{t=1}^T$, which is truncated into K segments, each of length L . This truncation is equivalent to using a rectangle window, which creates artificial discontinuities at the boundaries. This window has the form (Brockwell and Davis (2016)):

$$w(t) = \begin{cases} 1 & \text{if } |x| \leq 1 \\ 0 & \text{otherwise} \end{cases} \tag{36}$$

and the spectral window (the normalized squared Fourier transform of the taper) is given by the Dirichlet kernel:

$$W_{\text{rect}}(\omega) \propto \frac{1}{2\pi} \frac{\sin((r + \frac{1}{2})\omega)}{\sin(\omega/2)}, \quad r = \frac{L-1}{2} \tag{37}$$

With this window, $W(\omega)$ is negative at certain ω , which leads to a negative estimate of the spectral density at some frequencies ω . To mitigate this, we introduce multiple windows: Barlett,

Hanning and Blackman, which are covered in Harris (1978). Our series has a general form:

$$Y_t^{(k)} = w(t)X_t^{(k)} \quad (38)$$

All taper windows reduce the weights at the segment endpoints, with $w(0) \approx 0$ and $w(L-1) \approx 0$. Therefore, to avoid loss of information, we are going to truncate the stationary series into overlapping segments. The overlap portions ensure that observations near the boundary of one block appear near the center of another, where the taper weights are close to 1. It is controlled by the shift D , which for 50% overlap, $D = L/2$. If the shift is D , then the k -th segment is

$$X_t^{(k)} = X_{t+(k-1)D} \quad (39)$$

Using the taper windows, the series is now defined as

$$Y_t^{(k)} = w(t)X_t^{(k)} \quad (40)$$

where $w(t)$ represents the time-domain taper window.

We then proceed to compute the tapered periodogram of each segment using the window-power normalization, which is taper window specific.

$$I_k(\omega) = \frac{1}{2\pi L U_w} \left| \sum_{t=0}^{L-1} Y_t^{(k)} e^{-i\omega t} \right|^2, \quad (41)$$

where

$$U_w = \frac{1}{L} \sum_{t=0}^{L-1} w(t)^2 \quad (42)$$

is the window-power normalization constant, ensuring that the estimator is correctly scaled.

The Welch spectral density estimator is obtained by averaging the tapered periodograms over all K segments:

$$\hat{f}_{\text{Welch}}(\omega) = \frac{1}{K} \sum_{k=1}^K I_k(\omega). \quad (43)$$

Taper windows like Hanning and Blackman can be written as linear combinations of cosines, and cosines correspond to frequency shifts. Therefore, their spectral windows can be written as weighted combinations of shifted rectangular spectral windows from Eqn. 37. The Barlett window is constructed as a triangular, not a combination of cosines, hence it is the convolution of two rectangular windows. These spectral windows $W(\omega)$ will be used later to compare the speed of convergence to the true $f(\omega)$, since even though the estimates using these taper windows are consistent, they are biased. **Hanning (Hann) Window.**

$$w(t) = 0.5 \left(1 - \cos \left(\frac{2\pi t}{L-1} \right) \right), \quad t = 1, \dots, L-1 \quad (44)$$

$$W_{\text{Hann}}(\omega) = \frac{1}{2} W_{\text{rect}}(\omega) - \frac{1}{4} W_{\text{rect}}(\omega + \frac{2\pi}{L-1}) - \frac{1}{4} W_{\text{rect}}(\omega - \frac{2\pi}{L-1})$$

where $W_{rect}(\omega)$ is the rectangle spectral window from Eqn. 37.

Blackman Window.

$$\begin{aligned} w_{\text{Blackman}}(t) &= 0.42 - 0.5 \cos\left(\frac{2\pi t}{L-1}\right) + 0.08 \cos\left(\frac{4\pi t}{L-1}\right), \quad t = 0, \dots, L-1. \\ W_{\text{Blackman}}(\omega) &= 0.42 W_{\text{rect}}(\omega) - 0.25 W_{\text{rect}}\left(\omega + \frac{2\pi}{L-1}\right) - 0.25 W_{\text{rect}}\left(\omega - \frac{2\pi}{L-1}\right) \\ &\quad + 0.04 W_{\text{rect}}\left(\omega + \frac{4\pi}{L-1}\right) + 0.04 W_{\text{rect}}\left(\omega - \frac{4\pi}{L-1}\right) \end{aligned} \quad (45)$$

Bartlett Window.

$$\begin{aligned} w_{\text{Bartlett}}(t) &= (w_{\text{rect}} * w_{\text{rect}})(t) = \begin{cases} \frac{2t}{L-1}, & 0 \leq t \leq \frac{L-1}{2}, \\ 2 - \frac{2t}{L-1}, & \frac{L-1}{2} \leq t \leq L-1. \end{cases} \\ W_{\text{Bartlett}}(\omega) &= |W_{\text{rect}}(\omega)|^2 = \left(\frac{\sin(\omega L/2)}{\sin(\omega/2)}\right)^2 \end{aligned} \quad (46)$$

2.5. Simulation Study: Spectral Density Estimation via Welch’s Method

To illustrate the nonparametric procedure, we implement Welch’s method using Bartlett, Hanning and Blackman windows across different segment lengths $L \in \{256, 512, 1024\}$. We first apply the taper to each truncated segment. For each choice of window and segment length, we first estimate the sample autocovariance function for the truncated segments and then compute the corresponding windowed periodograms—i.e., the discrete Fourier transforms of the tapered ACVF estimates—using the appropriate normalization constant U_w . The Welch spectral estimate $\hat{F}_{\text{Welch}}$ is obtained by averaging these windowed periodograms across all segments. Figure 5 provides a visual comparison of these window-based estimators against the true spectral density, allowing us to assess how window choice and segment length affect the quality and smoothness of the estimates.

We assess estimation accuracy using the KL divergence, which quantifies the difference between the estimated spectral density and the true density by calculating the expected log-likelihood loss when estimating. We choose to use KL divergence since it penalizes for deviations in both shape and magnitude of the spectrum. By computing the KL divergence for all combinations of taper windows and segment lengths, we identify the window–length configuration that provides the closest nonparametric approximation to the true spectrum.

$$D_{KL}(f||\hat{F}) \approx \sum_{k=1}^K f(\omega_k) \log \left(\frac{f(\omega_k)}{\hat{F}(\omega_k)} \right) \Delta\omega. \quad (47)$$

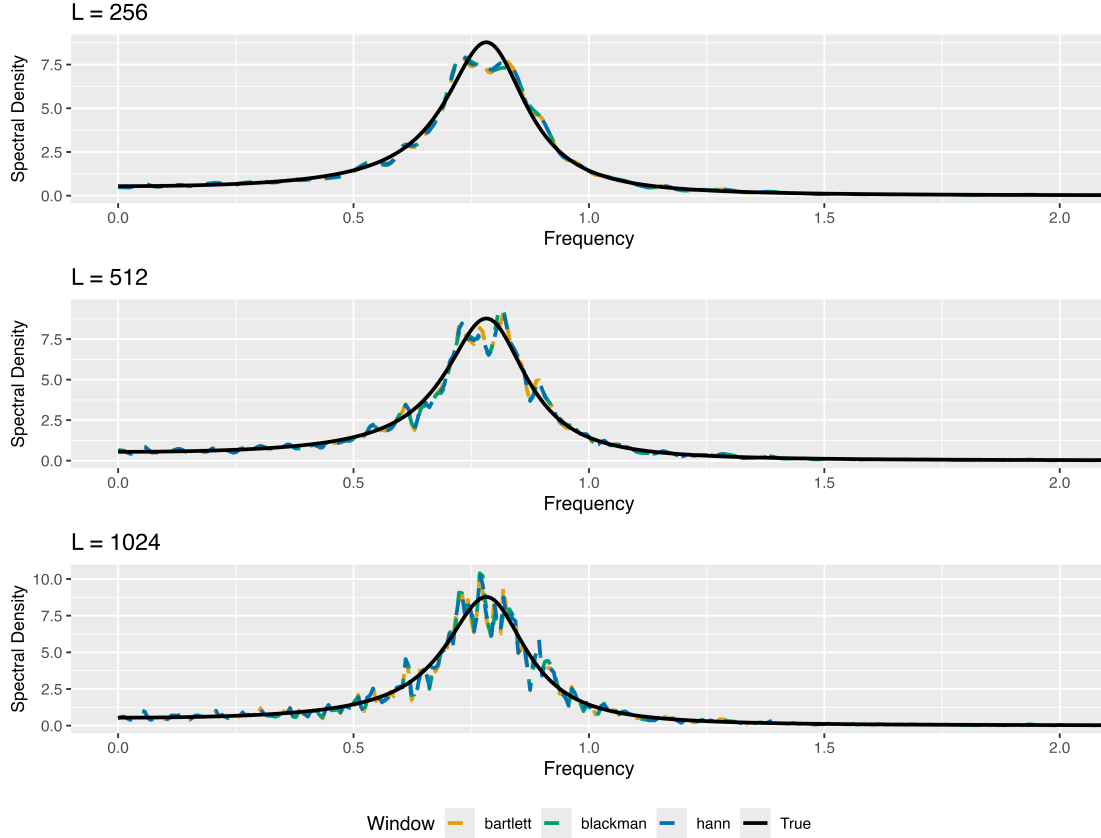


Fig. 5.— **Comparison of Spectral Density Estimates.** This figure displays the estimated spectral densities using 3 different taper windows across 3 segment lengths ($L = 256, 512, 1024$). The shorter segments appear smoother, and the three windows produce similar estimates.

Our result in Table 1 indicates that the Hanning window with segment length 256 yields the closest estimated spectral density.

Table 1: This table displays the KL divergence of each of the taper window method with various segment length and the true spectral density found from our simulation study. Based on this result, Hanning window with length 256 seems to be outperforming all other windows and lengths.

Segment Length	Hann	Bartlett	Blackman
256	0.056	0.067	0.059
512	0.085	0.092	0.093
1024	0.121	0.122	0.121

Hypothesis Test

We conduct this hypothesis test to assess whether the estimated spectral density is statistically consistent with the true spectral density implied by the underlying AR(2) model. The classical one-sample Kolmogorov–Smirnov (KS) test compares an empirical CDF with a fully specified theoretical CDF.

$$\begin{aligned} H_0 : \hat{F}(\omega) &= F(\omega) \\ H_A : \hat{F}(\omega) &\neq F(\omega) \end{aligned} \tag{48}$$

Under the assumption that the n observations are i.i.d., the KS statistic is:

$$D = \sup_x |F_n(x) - F(x)| \tag{49}$$

which has a known Kolmogorov distribution, from which standard critical values and p-values are derived.

However, this theoretical justification relies on i.i.d. observations and the classical empirical–CDF framework. Because the KS test assumes i.i.d. data and a classical empirical–CDF setting, the standard Kolmogorov critical values are not valid for our spectral–density comparison problem due to correlation in temporal data. Consequently, the sampling distribution of the KS-type statistic is unknown. To obtain a valid test, we employ a bootstrap procedure under the null model. We generate new time-series replicates by sampling the (approximately Gaussian) residuals and reconstructing a stationary AR(2) process that would have arisen if the null hypothesis were true, thereby preserving the temporal dependence structure. For each bootstrap sample b , we apply the same Welch estimation pipeline, compute the corresponding spectral CDF $\hat{F}^{(b)}(\omega) = \int_0^\omega \hat{F}_{Welch}(\lambda)^{(b)} d\lambda$, and calculate the KS-type statistic $D^{(b)} = \sup_\omega |\hat{F}^{(b)}(\omega) - F(\omega)|$ for each bootstrap sample. The collection $D^{(b)}_{b=1}^B$ provides an empirical approximation to the null distribution of the test statistic. The bootstrap p-value given by the proportion of bootstrap statistics exceeding the observed statistic correctly accounts for temporal dependence as well as all transformations involved in spectral estimation, yielding a valid test for equality of spectral densities:

$$\hat{p} = \frac{1}{B} \sum_{b=1}^B \mathbb{1}_{D^{(b)} > D_{obs}}, \tag{50}$$

where B is the number of bootstrap samples, $D^{(b)}$ is the KS test statistic of that bootstrap sample and D_{obs} is the observed test statistics.

Based on the results from Table 1, we conduct the hypothesis test for the spectral density computed using Hanning taper window with length $L = 256$.

The observed KS test statistics between the estimated spectral distribution $\hat{F}(\omega)$ and the true spectral distribution $F(\omega)$ is $D_{obs} = 0.017887$.

Under the null hypothesis H_0 , we simulate data from the AR(2) model with parameters $\phi = (1.27, -0.81)$, $\sigma = 1$, and compute the KS-type statistic for each bootstrap replication. This yields

the empirical null distribution of D . We obtain the bootstrap p-value

$$\hat{p} = \mathbb{P}(D^{(b)} \geq D_{\text{obs}} \mid H_0) = 0.584. \quad (51)$$

Thus, if H_0 is true, there is a 58.2% chance of obtaining a value of D at least as large as the observed statistic. At significance level $\alpha = 0.05$, we have $\hat{p} = 0.582 \geq 0.05$, therefore we fail to reject the null hypothesis. This indicates that the estimated spectral density is consistent with the true spectral density and that the estimation procedure produces reasonable results.

3. Conclusion

This report has investigated both parametric and nonparametric approaches to spectral density estimation through theoretical development and comprehensive simulation studies. Our investigation establishes the fundamental approximation theorem demonstrating that any stationary process spectral density can be arbitrarily closely approximated by an $\text{AR}(p)$ spectral density under appropriate conditions. The parametric approach, examined through an $\text{AR}(2)$ model specification, demonstrates strong performance when the model structure aligns with the underlying data-generating process. The CvM goodness-of-fit test fails to reject the null hypothesis of distributional equivalence between the theoretical and empirical spectral distribution functions, confirming that the correctly specified parametric model provides an efficient and parsimonious representation of the spectral characteristics. However, our model misspecification analysis reveals the critical vulnerability of parametric methods: models employing incorrect functional forms, such as imposing MA structure when the true process follows an AR specification, or insufficient model orders results in larger KL divergence. This consequence stems from the fundamental constraint that under-specified or misspecified parametric models lack the structural flexibility to accurately reconstruct complex spectral densities. In contrast, the nonparametric approach demonstrates robustness to model uncertainty. While the initial circulant matrix implementation proved inconsistent for direct spectral density estimation, Welch’s method improves the estimates without imposing restrictive parametric assumptions. The KS test corroborates the reliability of this optimally-tuned nonparametric estimator, demonstrating its ability to capture features of the spectrum that simpler parametric models may fail to reproduce. These complementary findings shed light on the fundamental trade-off governing spectral density estimation methodology. Parametric methods tend to produce better results than classical nonparametric methods when the data length is relatively short but are subject to error from model misspecification. When the true spectral density exhibits complex structure—involving multiple peaks, irregular patterns, or features not readily captured by low-order AR or ARMA models—parametric approaches often fail to provide adequate approximations, regardless of estimation precision. Conversely, nonparametric methods may exhibit lower efficiency under correct parametric specification, yet they provide essential robustness when the true spectrum is too complex for low-order parametric models to capture. This robustness proves particularly valuable in exploratory analysis or applications where prior knowledge of the spectral structure is limited or unreliable. For example, macroeconomics times series often has structural

breaks that violates the assumptions of the parametric models, in this case, nonparametric methods provide estimates without requiring the data to conform to a specific theoretical model. An interesting direction for future work is to explore the semiparametric spectral estimators, such as the local Whittle or hybrid AR–nonparametric smoothing methods, which combine the structural interpretability of parametric models with the flexibility of nonparametric estimators. Such approaches may offer improved robustness in settings where the spectrum exhibits both smooth long-run features and localized irregularities that are not well captured by either method alone.

4. Appendix

For the choice of δ in the proof: From Eqn. 14, we want to restrict the bound such that $\frac{\delta}{|XY|}$ does not get too large, which means we want $|XY|$ to not get too small, therefore we will find the lower bound for it. Since we have $Y > \frac{1}{M}$, and $X > \frac{1}{M} - \delta$, we can consider the choice of δ to be at most $\frac{1}{2M}$.

Additionally, with the ϵ - δ style proof, we want

$$\left| \frac{1}{X} - \frac{1}{Y} \right| \leq \frac{\epsilon}{2}. \quad (52)$$

From Eqn. 14, we also have

$$\left| \frac{1}{X} - \frac{1}{Y} \right| \leq \frac{\delta}{|XY|}. \quad (53)$$

Therefore, to satisfy Eqn. 52, we must choose δ such that

$$\frac{\delta}{|XY|} \leq \frac{\epsilon}{2} \iff \delta \leq \frac{\epsilon}{2}|XY| \quad (54)$$

From Eqn. 16

$$|XY| > \frac{1}{2M^2}, \quad (55)$$

it follows that

$$\frac{\epsilon}{2}|XY| \geq \frac{\epsilon}{4M^2} \quad (56)$$

Therefore, combining with the condition that $\delta \geq \frac{1}{2M}$, we choose

$$\delta = \min \left(\frac{1}{2M}, \frac{\epsilon}{4M^2} \right)$$

to guarantee that :

$$|[A|\phi(e^{-i\lambda})|^2]^{-1} - f_p(\lambda)| \leq \frac{\epsilon}{2}. \quad (57)$$

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